N個の自由粒子　$H = \sum^{3N} \frac{p_i^2}{2m}$

状態数 $J(E)$ = 半径 $\sqrt{2mE}$ の $3N$ 次元球の体積 $/ \left(\frac{h}{L}\right)^{3N} N!$　← 粒子が区別できない

$$= \frac{V^N}{h^{3N} N!} \cdot \frac{\pi^{\frac{3N}{2}}}{\Gamma\left(\frac{3N}{2}+1\right)} (2mE)^{\frac{3N}{2}} = \left(\frac{eV}{h^3 N}\right)^N \left(\frac{2e}{3} 2\pi m \frac{E}{N}\right)^{\frac{3N}{2}}$$

状態密度 $\Omega(E) = \frac{dJ}{dE} \sim J$

$$S = k \ln \Omega(E) = Nk\left\{\ln \frac{V}{N} + \frac{3}{2} \ln \frac{2E}{3N} + \ln \frac{(2\pi m)^{\frac{3}{2}} e^{\frac{5}{2}}}{h^3}\right\}$$

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{3Nk}{2E} \qquad C_v = \frac{\partial E}{\partial T} = \frac{3}{2} Nk$$

$$\frac{P}{T} = \frac{\partial S}{\partial V} = \frac{Nk}{V} \qquad PV = NkT$$

分配関数　$Z = \frac{1}{h^{3N} N!} \int e^{-\frac{H}{kT}} dp\, dq = \frac{V^N}{h^{3N} N!} (2\pi m kT)^{\frac{3N}{2}} \approx \left(\frac{eV}{N}\right)^N \underbrace{\left(\frac{mkT}{2\pi\hbar^2}\right)^{\frac{3N}{2}}}_{\lambda_T^{-2}}$

$$F = -kT \ln Z = -NkT \ln \frac{eV}{h^3 N} (2\pi m kT)^{\frac{3}{2}}$$

$$S = -\frac{\partial F}{\partial T} = Nk \ln Z + \frac{3}{2} Nk = Nk \ln \frac{e^{\frac{5}{2}} V}{h^3 N} (2\pi m kT)^{\frac{3}{2}}$$

$$U = F + TS = kT^2 \frac{\partial}{\partial T} \ln Z = \frac{3}{2} NkT \qquad C_v = \frac{\partial U}{\partial T} = \frac{3}{2} Nk$$

$$P = -\frac{\partial F}{\partial V} = \frac{NkT}{V} \qquad PV = NkT$$

$$H = U + PV = \frac{5}{2} NkT \qquad C_p = \frac{\partial H}{\partial T} = \frac{5}{2} Nk$$

熱的 de Broglie 波

$kT \sim \frac{\langle p \rangle^2}{2m}$

$\lambda_T \sim \frac{h}{\langle p \rangle}$

n次元球の体積　$V = C(n) r^n$　表面積 $S = n C(n) r^{n-1}$

$$\int e^{-a(x_1^2 + \cdots)} dx_1 \cdots = \int e^{-ar^2} n C(n) r^{n-1} dr = \frac{n}{2a} C(n) \int e^{-t} \left(\frac{t}{a}\right)^{\frac{n}{2}-1} dt = \frac{C(n)}{a^{\frac{n}{2}}} \frac{n}{2} \Gamma\left(\frac{n}{2}\right)$$

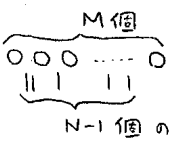
$$\left(\frac{\pi}{a}\right)^{\frac{n}{2}} \qquad = \frac{C(n)}{a^{\frac{n}{2}}} \Gamma\left(\frac{n}{2}+1\right)$$

$$\therefore C(n) = \pi^{\frac{n}{2}} / \Gamma\left(\frac{n}{2}+1\right)$$

N個の振動子　　$\varepsilon_i = \hbar\omega\left(n_i + \frac{1}{2}\right)$

$$E = \frac{N}{2}\hbar\omega + M\hbar\omega \qquad M = n_1 + n_2 + \cdots + n_N$$

状態数 $W = {}_{M+N-1}C_{N-1} = \dfrac{(M+N-1)!}{M!\,(N-1)!}$



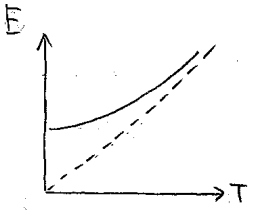
$$S = k\ln W = k\left[(M+N)\ln(M+N) - M\ln M - N\ln N\right] \qquad (M, N \gg 1)$$

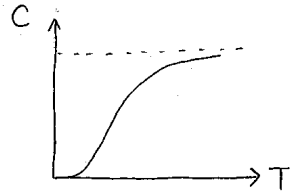
$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{k}{\hbar\omega}\ln\frac{M+N}{M} = \frac{k}{\hbar\omega}\ln\frac{E+\frac{N}{2}\hbar\omega}{E-\frac{N}{2}\hbar\omega}$$

$$\frac{E+\frac{N}{2}\hbar\omega}{E-\frac{N}{2}\hbar\omega} = e^{\frac{\hbar\omega}{kT}}$$

$$E = N\left\{\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\frac{\hbar\omega}{kT}} - 1}\right\}$$

$$C_v = \frac{\partial E}{\partial T} = Nk\left(\frac{\frac{\hbar\omega}{kT}}{2\sinh\left(\frac{\hbar\omega}{2kT}\right)}\right)^2$$





分配関数 $Z = \sum e^{-\frac{E}{kT}} = \left[\sum e^{-\frac{\hbar\omega}{kT}\left(n+\frac{1}{2}\right)}\right]^N = \left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right]^{-N}$

$$F = -kT\ln Z = NkT\ln\left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right]$$

$$U = F + TS = -NkT^2\frac{\partial}{\partial T}\ln\left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right] = \frac{N}{2}\hbar\omega\Big/\tanh\left(\frac{\hbar\omega}{2kT}\right) = N\left\{\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\frac{\hbar\omega}{kT}} - 1}\right\}$$

$$C_v = \frac{\partial U}{\partial T} = Nk\left(\frac{\frac{\hbar\omega}{2kT}}{\sinh\left(\frac{\hbar\omega}{2kT}\right)}\right)^2$$

古典的　$H = \dfrac{p^2}{2m} + \dfrac{m\omega^2}{2}q^2$

$$Z = \frac{1}{h^{3N}N!}\int e^{-\frac{H}{kT}}\,dp\,dq = \left(\frac{e}{N}\right)^N\left(\frac{2\pi kT}{\hbar\omega}\right)^{3N}$$

$$F = -3NkT\ln\left(\frac{e}{N}\right)^{\frac{1}{3}}\left(\frac{2\pi kT}{\hbar\omega}\right)$$

$$U = 3NkT \qquad C_v = 3Nk$$

熱輻射

$V = L^3$　　$\vec{k} = (k_x, k_y, k_z) \sim (k_x + dk_x, k_y + dk_y, k_z + dk_z)$

$\frac{2\pi}{L}(n_x, n_y, n_z)$　　この領域に存在する基準振動の個数

$\omega = ck$

$$\frac{V}{(2\pi)^3} dk_x dk_y dk_z \times 2 = \frac{V}{(2\pi)^3} 4\pi k^2 dk \times 2 = D(k)dk$$

↑ 偏りの自由度

$$D(\omega) = D(k)\frac{dk}{d\omega} = \frac{V\omega^2}{\pi^2 c^3} d\omega$$

$\varepsilon = \frac{1}{2}\hbar\omega + \langle n\rangle \hbar\omega$　　$\langle n\rangle = \frac{1}{e^{\frac{\hbar\omega}{kT}} - 1}$　　Bose 分布　$\mu = 0$

$$U = U_0 + \int_0^\infty d\omega\, D(\omega)\,\hbar\omega\langle n\rangle = U_0 + \frac{V\hbar}{\pi^2 c^3}\int_0^\infty d\omega \frac{\omega^3}{e^{\frac{\hbar\omega}{kT}} - 1}$$

$$U(\omega) = \frac{V\hbar}{\pi^2 c^3} \cdot \frac{\omega^3}{e^{\frac{\hbar\omega}{kT}} - 1}$$

$$= U_0 + V\frac{(kT)^4}{\pi^2 c^3 \hbar^3}\int_0^\infty \frac{x^3 dx}{e^x - 1} = U_0 + V\frac{\pi^2 k^4}{15 c^3 \hbar^3} T^4$$

$$\sum_{n=1}^\infty \int_0^\infty x^3 e^{-nx} dx = \sum_{n=1}^\infty \frac{1}{n^4}\int_0^\infty e^{-t} t^3 dt = \underbrace{P(4)}_{3!} \underbrace{\sum_{n=1}^\infty \frac{1}{n^4}}_{\zeta(4) = \frac{\pi^4}{90}} = \frac{\pi^4}{15}$$

Debye - model

$$D(\omega) = \begin{cases} \frac{V}{2\pi^2}\left(\frac{1}{c_l^3} + \frac{2}{c_t^3}\right)\omega^2 & \omega < \omega_D \\ 0 & \omega > \omega_D \end{cases}$$

$$3N = \int D(\omega)d\omega = \frac{V}{2\pi^2}\left(\frac{1}{c_l^3} + \frac{2}{c_t^3}\right)\frac{\omega_D^3}{3}$$

$$U = \int_0^\infty \varepsilon(\omega) D(\omega) d\omega = \int_0^{\omega_D} \frac{9N}{\omega_D^3}\omega^2\left(\frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\frac{\hbar\omega}{kT}} - 1}\right) d\omega$$

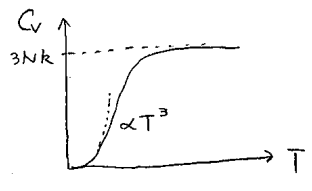
$\frac{\hbar\omega_D}{kT} = \frac{\theta_D}{T}$　Debye 温度

$$= 9NkT\left(\frac{T}{\theta_D}\right)^3 \int_0^{\frac{\theta_D}{T}} \left(\frac{1}{2} + \frac{1}{e^x - 1}\right) x^3 dx = \frac{9N}{8} k\theta_D + 3NkT\,D\left(\frac{\theta_D}{T}\right)$$

$$D(y) = \frac{3}{y^3}\int_0^y \frac{x^3}{e^x - 1} dx \qquad D'(y) = \frac{3}{e^y - 1} - \frac{3}{y} D(y)$$

$$C_V = \frac{\partial U}{\partial T} = 3Nk\left(D - \frac{\theta_D}{T} D'\right) = 3Nk\left(4D\left(\frac{\theta_D}{T}\right) - \frac{3\theta_D/T}{e^{\theta_D/T} - 1}\right)$$

$$= \begin{cases} 3Nk\left(\frac{4\pi^4}{5}\left(\frac{T}{\theta_D}\right)^3 - 3\frac{\theta_D}{T} e^{-\frac{\theta_D}{T}} \cdots\right) & T \ll \theta_D \\ 3Nk \end{cases}$$



N個の2準位系　$-\varepsilon_0, \varepsilon_0$

$$E = N_+\varepsilon_0 - N_-\varepsilon_0 = M\varepsilon_0 \qquad M = N_+ - N_- \qquad N = N_+ + N_-$$

$$W = \frac{N!}{N_+!\,N_-!} = \frac{N!}{\left[\frac{N-M}{2}\right]!\left[\frac{N+M}{2}\right]!}$$

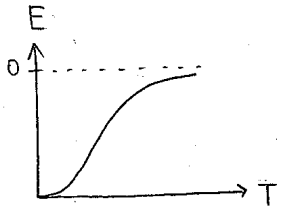
$$S = k\ln W = k\left[N\ln N - \frac{N-M}{2}\ln\frac{N-M}{2} - \frac{N+M}{2}\ln\frac{N+M}{2}\right]$$

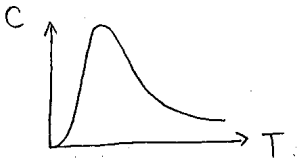
$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{k}{2\varepsilon_0}\ln\frac{N-M}{N+M} \qquad \frac{N_-}{N_+} = e^{\frac{2\varepsilon_0}{kT}} \qquad \frac{N_+}{N} = \frac{e^{-\varepsilon_0/kT}}{e^{\varepsilon_0/kT}+e^{-\varepsilon_0/kT}}$$

$$\frac{N_-}{N} = \frac{e^{\varepsilon_0/kT}}{e^{\varepsilon_0/kT}+e^{-\varepsilon_0/kT}}$$

$$E = -N\varepsilon_0\tanh\left(\frac{\varepsilon_0}{kT}\right)$$

$$C = Nk\left(\frac{\frac{\varepsilon_0}{kT}}{\cosh\left(\frac{\varepsilon_0}{kT}\right)}\right)^2$$





Schottky 型比熱

分配関数 $Z = \left[e^{\varepsilon_0/kT} + e^{-\varepsilon_0/kT}\right]^N = \left[2\cosh\left(\frac{\varepsilon_0}{kT}\right)\right]^N$

$$F = -NkT\ln\left[2\cosh\left(\frac{\varepsilon_0}{kT}\right)\right]$$

$$U = kT^2\frac{\partial}{\partial T}\ln Z = -N\varepsilon_0\tanh\left(\frac{\varepsilon_0}{kT}\right)$$

Zeeman効果　$\varepsilon_0 = \mu H$

磁気モーメント $M = -\frac{\partial F}{\partial H} = NkT\frac{\partial}{\partial H}\ln\left[2\cosh\left(\frac{\mu H}{kT}\right)\right] = N\mu\tanh\left(\frac{\mu H}{kT}\right)$

分布関数　C_j 個の量子状態　N_j 個の粒子　$\gg 1$　　$N=\sum_j N_j$　$E=\sum_j \varepsilon_j N_j$

$$W_{F.D}=\prod_j {}_{C_j}C_{N_j}=\prod_j \frac{C_j!}{(C_j-N_j)!\,N_j!}=\prod_j \left(\frac{C_j}{C_j-N_j}\right)^{C_j}\left(\frac{C_j-N_j}{N_j}\right)^{N_j}$$

$$W_{B.E}=\prod_j {}_{N_j+C_j-1}C_{C_j-1}=\prod_j \frac{(N_j+C_j-1)!}{N!\,(C_j-1)!}=\prod_j \left(\frac{C_j+N_j}{C_j}\right)^{C_j}\left(\frac{C_j+N_j}{N_j}\right)^{N_j}$$

(i) ミクロカノニカル

$$\left.\begin{aligned}\delta S&=k\,\delta \ln W=k\sum_j\{\ln(C_j\mp N_j)-\ln N_j\}\,\delta N_j\\ \delta S&=\frac{\partial S}{\partial N}\delta N+\frac{\partial S}{\partial E}\delta E=-\frac{\mu}{T}\sum_j \delta N_j+\frac{1}{T}\sum_j \varepsilon_j \delta N_j\end{aligned}\right\}\quad \frac{C_j\mp N_j}{N_j}=e^{\frac{\varepsilon_j-\mu}{kT}}$$

1量子状態当りの平均粒子数　$n_j=\dfrac{N_j}{C_j}=\dfrac{1}{e^{\frac{\varepsilon_j-\mu}{kT}}\pm 1}$

(ii) カノニカル

$$Z=W\cdot e^{-\frac{E}{kT}}\qquad F=-kT\ln Z=E-kT\ln W$$

$$\left.\begin{aligned}\delta F&=\sum_j \varepsilon_j\,\delta N_j-kT\sum_j\{\ln(C_j\mp N_j)-\ln N_j\}\,\delta N_j\\ \delta F&=\frac{\partial F}{\partial N}\delta N=\mu\sum_j \delta N_j\end{aligned}\right\}\quad \frac{\varepsilon_j-\mu}{kT}=\ln\frac{C_j\mp N_j}{N_j}$$

(iii) グランドカノニカル

$$\Xi=\sum_{N=0}^{\infty}\sum_E e^{-\frac{E-\mu N}{kT}}=\sum_{N=0}^{\infty}\sum_E e^{\sum_j \frac{\mu-\varepsilon_j}{kT}N_j}=\sum_{N_1}\cdots\sum_{N_N}\prod_j\left[e^{\frac{\mu-\varepsilon_j}{kT}}\right]^{N_j}=\prod_j\sum_{N_j}\left[e^{\frac{\mu-\varepsilon_j}{kT}}\right]^{N_j}$$

$$\Xi_{F.D}=\prod_j\sum_{N_j=0}^{1}\left[e^{\frac{\mu-\varepsilon_j}{kT}}\right]^{N_j}=\prod_j\left(e^{\frac{\mu-\varepsilon_j}{kT}}+1\right)$$

$$\Xi_{B.E}=\prod_j\sum_{N_j=0}^{\infty}\left[e^{\frac{\mu-\varepsilon_j}{kT}}\right]^{N_j}=\prod_j\frac{1}{1-e^{\frac{\mu-\varepsilon_j}{kT}}}$$

$$\mathcal{Z} = \sum_n e^{\beta(\mu-\varepsilon)n} = \left(1 \pm e^{\beta(\mu-\varepsilon)}\right)^{\mp 1}$$

+ : Fermion

− : Boson

$$f(\varepsilon) = \langle n \rangle = \frac{1}{\beta}\frac{\partial}{\partial \mu}\ln \mathcal{Z} = \frac{1}{e^{\beta(\varepsilon-\mu)} \pm 1}$$

$$D(\varepsilon) = \frac{gV}{h^3}\frac{\partial}{\partial p}\int_0^p 4\pi p^2 dp \,, \qquad W(\varepsilon) = \int_0^\varepsilon 3p(\varepsilon) D(\varepsilon) d\varepsilon = \frac{gV}{h^3}\cdot\frac{4\pi}{3}p^3$$

$$\Pi = \prod_i^3 \mathcal{Z}^{3p(\varepsilon)D\varepsilon} \qquad \ln \Pi = \int 3p(\varepsilon) D(\varepsilon) \ln \mathcal{Z}\, d\varepsilon$$

$$\Omega = -PV = -\frac{1}{\beta}\ln \Pi = -\frac{1}{\beta}\left[\ln \mathcal{Z}\right]_0^\infty + \frac{1}{\beta}\int W(\varepsilon)\frac{\partial}{\partial \varepsilon}\ln \mathcal{Z}\, d\varepsilon = -\int W(\varepsilon) f(\varepsilon) d\varepsilon$$

$$N = -\left(\frac{\partial \Omega}{\partial \nu}\right)_{T} = \frac{1}{\beta}\frac{\partial}{\partial \nu}\ln \Pi = \int 3p(\varepsilon) D(\varepsilon) \frac{1}{\beta}\frac{\partial}{\partial \varepsilon}\ln \mathcal{Z}\, d\varepsilon = \int 3p(\varepsilon) f(\varepsilon) D(\varepsilon) d\varepsilon$$

$$E = \mu N - \frac{\partial}{\partial \beta}\ln \Pi = \int D(\varepsilon)\left(\frac{\varepsilon}{\varepsilon}\frac{\beta}{3} + \frac{\nu}{\varepsilon}\frac{\beta}{\nu}\right) 3p \ln \mathcal{Z}\, d\varepsilon = N\mu + \int D(\varepsilon)\frac{\beta}{3}\frac{\partial}{\partial \varepsilon} 3p \ln \mathcal{Z}\, d\varepsilon$$

$$P = -\left(\frac{\partial \Omega}{\partial V}\right)_{T,\mu} = \frac{1}{\beta}\frac{\partial}{\partial \varepsilon}\ln \Pi = \frac{1}{\beta}\int \frac{\partial D(\varepsilon)}{\partial \varepsilon} 3p \ln \mathcal{Z}\, d\varepsilon = -\frac{\Omega}{V} = \frac{1}{V}\int W(\varepsilon) f(\varepsilon) d\varepsilon$$

$$E = -\int D(\varepsilon)\frac{\beta}{3} 3p \ln \mathcal{Z}\, d\varepsilon = \int 3p(\varepsilon) f 3(\varepsilon) D(\varepsilon)$$

$$\varepsilon^2 - c^2p^2 = m^2c^4$$

非相対論 $\varepsilon = \frac{p^2}{2m}$ $\quad \frac{3W(\varepsilon)}{3D(\varepsilon)} = \frac{2}{\varepsilon}$ $\quad PV = \frac{2}{3}E$ $\quad W(\varepsilon) = \frac{4\pi gV}{3h^3}(2m\varepsilon)^{\frac{3}{2}}$ $\quad P = \frac{1}{5m}\left(\frac{3h^3}{4\pi g}\right)^{\frac{2}{3}} n^{\frac{5}{3}}$

超相対論 $\varepsilon = cp$ $\quad \frac{3W(\varepsilon)}{3D(\varepsilon)} = 3$ $\quad PV = \frac{1}{3}E$ $\quad W(\varepsilon) = \frac{4\pi gV}{h^3}\left(\frac{\varepsilon}{c}\right)^3$ $\quad P = \frac{c}{4}\left(\frac{3h^3}{4\pi g}\right)^{\frac{1}{3}} n^{\frac{4}{3}}$

完全縮退 Fermion $T=0$ $\quad f(\varepsilon) = \theta(\mu-\varepsilon)$

$$N = \int_0^\mu D(\varepsilon) 3p\, d\varepsilon = W(\mu)$$

$$E = \int_0^\mu D(\varepsilon) 3p\, \varepsilon\, d\varepsilon = W(\mu)\mu - \int_0^\mu W(\varepsilon) d\varepsilon = N\mu - PV$$

$$P = \frac{1}{V}\int_0^\mu W(\varepsilon) 3p\, d\varepsilon$$

$$g\,\frac{V\,\frac{4\pi}{3}P_F^3}{h^3} = N$$

(N.R.) $\quad \varepsilon_F = \frac{P_F^2}{2m} = \frac{1}{2m}\left(\frac{3h^3}{4\pi g}\right)^{\frac{2}{3}} n^{\frac{2}{3}}$

(E.R.) $\quad \varepsilon_F = cP_F = c\left(\frac{3h^3}{4\pi g}\right)^{\frac{1}{3}} n^{\frac{1}{3}}$

$$p = \frac{\partial(\varepsilon/\rho)}{\partial(1/\rho)} = \frac{\partial(\varepsilon/n)}{\partial(1/n)} \stackrel{?}{=} \frac{\partial \varepsilon_F}{\partial\left(\frac{1}{n}\right)}$$

$$L = \frac{m}{2}\dot{x}^2 - e\phi + e\dot{x}\cdot A$$

$$P = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + eA$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = m\ddot{x} + e\frac{dA}{dt} + e\frac{\partial \phi}{\partial x} - e\frac{\partial}{\partial x}(\dot{x}\cdot A)$$

$$= m\ddot{x} + e\frac{\partial \phi}{\partial x} + e\frac{\partial A}{\partial t} + e\left(\dot{x}\cdot\frac{\partial}{\partial x}\right)A - e\frac{\partial}{\partial x}(\dot{x}\cdot A) \qquad \leftarrow \frac{d}{dt} = \frac{\partial}{\partial t} + \left(\dot{x}\cdot\frac{\partial}{\partial x}\right)$$

$$= m\ddot{x} + e\left(\frac{\partial \phi}{\partial x} + \frac{\partial A}{\partial t}\right) - e\,\dot{x}\times(\nabla\times A)$$

$$\therefore \quad m\ddot{x} = e(E + \dot{x}\times B)$$

$$H = P\cdot\dot{x} - L = \frac{m}{2}\dot{x}^2 + e\phi = \frac{(P-eA)^2}{2m} + e\phi$$

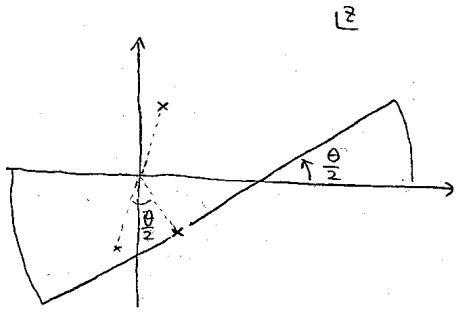
$$\int_{-\infty}^{\infty} e^{+Ax^2+Bx+C}\,dx \qquad A,B,C \in \mathbb{C}$$

$$= \int_{-\infty}^{\infty} e^{-ae^{i\theta}(x-b)^2+C}\,dx \qquad a>0,\ \theta \in \mathbb{R},\ b,C \in \mathbb{C}$$

$$= \int_{-\infty}^{\infty} e^{-az^2} \cdot e^{C} \cdot e^{-\frac{i\theta}{2}}\,dz \qquad e^{\frac{i\theta}{2}}(x-b) = z$$

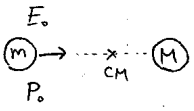
$$= e^{-\frac{i\theta}{2}} e^{C} \int_{-\infty}^{\infty} e^{-az^2}\,dz$$

$$= \sqrt{\frac{\pi}{a}}\, e^{-\frac{i\theta}{2}} e^{C}$$

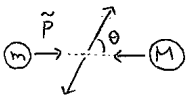


弾性散乱　　$E^2 - c^2p^2 = m^2c^4$

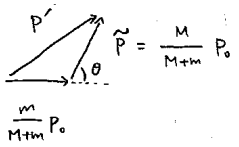
Ex)



CM)



i) N.R.　$E_0 = \frac{P_0^2}{2m}$　　$\tilde{P} = P_0 - \frac{m}{M+m}P_0 = \frac{M}{M+m}P_0$



$$P'^2 = \left(\frac{m}{M+m}P_0\right)^2 + \left(\frac{M}{M+m}P_0\right)^2 + 2\frac{mM}{(M+m)^2}P_0^2\cos\theta$$

$$\frac{E}{E_0} = \frac{M^2+m^2}{(M+m)^2} + \frac{2mM}{(M+m)^2}\cos\theta$$

ii) E.R.　$E_0 = cP_0$

ヒルベルト空間

$$\begin{cases} \hat{\delta}_k = \frac{1}{2\pi}\int dx\, e^{-ikx}\,\delta(x) = \frac{1}{2\pi} \\ \delta(x) = \int dk\, e^{ikx}\,\hat{\delta}_k = \frac{1}{2\pi}\int dk\, e^{ikx} \end{cases}$$

$$\langle x|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}}\int dp\,\langle p|\psi\rangle e^{\frac{i}{\hbar}px} \qquad \langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{\frac{i}{\hbar}px}$$

$$\langle p|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}}\int dx\,\langle x|\psi\rangle e^{-\frac{i}{\hbar}px}$$

$$\langle x'|x\rangle = \int dp\,\langle x'|p\rangle\langle p|x\rangle = \frac{1}{2\pi\hbar}\int dp\, e^{\frac{i}{\hbar}p(x'-x)} = \delta(x'-x)$$

ベーカー・ハウスドルフ の補助定理

$$\hat{F}(\lambda) = e^{\lambda \hat{S}} \hat{O} e^{-\lambda \hat{S}}$$

$$\hat{F}'(\lambda) = e^{\lambda \hat{S}} [\hat{S}, \hat{O}] e^{-\lambda \hat{S}}$$

$$\hat{F}''(\lambda) = e^{\lambda \hat{S}} [\hat{S}, [\hat{S}, \hat{O}]] e^{-\lambda \hat{S}}$$

$$e^{\lambda \hat{S}} \hat{O} e^{-\lambda \hat{S}} = \hat{O} + \lambda [\hat{S}, \hat{O}] + \frac{\lambda^2}{2!} [\hat{S}, [\hat{S}, \hat{O}]] + \cdots$$

$$e^{\frac{i}{\hbar} \hat{P} x'} \hat{x} e^{-\frac{i}{\hbar} \hat{P} x'} = \hat{x} + x'$$

$$e^{\frac{i}{\hbar} \hat{H} t} \hat{Q} e^{-\frac{i}{\hbar} \hat{H} t} = \hat{Q} + \frac{it}{\hbar} \underbrace{[\hat{H}, \hat{Q}]}_{\hbar\omega [\hat{P}^2, \hat{Q}] = -i\hat{P}} + \frac{1}{2} \left(\frac{it}{\hbar}\right)^2 \underbrace{[\hat{H}, [\hat{H}, \hat{Q}]]}_{-i\hbar\omega [\hat{H}, \hat{P}] = \hbar\omega [\hat{Q}^2, \hat{P}] = i\hat{Q}} + \cdots$$

$$= \hat{Q} + \omega t \hat{P} - \frac{1}{2} (\omega t)^2 \hat{Q} + \cdots$$

$$= \hat{Q} \cos \omega t + \hat{P} \sin \omega t$$

調和振動子

$$H = \hbar\omega(\hat{P}^2+\hat{Q}^2) = \hbar\omega(\underbrace{\hat{Q}-i\hat{P}}_{\hat{a}^\dagger})(\underbrace{\hat{Q}+i\hat{P}}_{\hat{a}}) - i\hbar\omega\underbrace{[\hat{Q},\hat{P}]}_{\frac{i}{2}}$$

$$[\hat{a},\hat{a}^\dagger] = 1$$

$$\underbrace{\hat{n}}_{\hat{a}^\dagger\hat{a}}|n\rangle = n|n\rangle \qquad |n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle$$

コヒーレント状態

$$\underbrace{\hat{a}}_{Q+\frac{1}{2}\frac{d}{dQ}}|z\rangle = z|z\rangle \qquad |z\rangle = e^{-\frac{|z|^2}{2}}\sum\frac{z^n}{\sqrt{n!}}|n\rangle \qquad \langle z|H|z\rangle = \hbar\omega\left(|z|^2+\tfrac{1}{2}\right)$$

$$\langle Q|z\rangle = \frac{1}{(2\pi)^{1/4}}e^{-(Q-z)^2} = \langle Q-z|0\rangle$$

$$|z(t)\rangle = e^{-iHt/\hbar}|z(0)\rangle = e^{-\frac{i\omega t}{2}}e^{-\frac{|z(t)|^2}{2}}\sum\frac{|z(t)|^n}{\sqrt{n!}}|n\rangle \qquad z(t) = z(0)e^{i\omega t}$$

角運動量　非アーベル群

$[J_i, J_j] = i\epsilon_{ijk} J_k \qquad J_\pm = J_x \pm i J_y$

$J^2 - J_z = \frac{1}{2}\{J_+, J_-\}$

$J_+ J_- = J^2 - J_z^2 + J_z$

$J^2 |J, M\rangle = J(J+1)|J, M\rangle$

$J_z |J, M\rangle = M |J, M\rangle$

シュレディンガー表示と ハイゼンベルグ表示

$$i\hbar \frac{\partial}{\partial t} |\alpha, t\rangle_S = H |\alpha, t\rangle_S$$

$$|\alpha, t\rangle_S = e^{-\frac{i}{\hbar}Ht} |\alpha\rangle_H$$

$$i\hbar \frac{dA^H(t)}{dt} = [A^H(t), H]$$

$$A^H(t) = e^{+\frac{i}{\hbar}Ht} A^S e^{-\frac{i}{\hbar}Ht}$$

$$\langle A(t) \rangle = {}_H\langle \alpha | A^H(t) | \alpha \rangle_H = {}_S\langle \alpha, t | A^S | \alpha, t \rangle_S$$

相互作用表示、 $H = H_0 + V$

$$|\alpha, t\rangle_S = e^{-\frac{i}{\hbar}H_0 t} |\alpha, t\rangle_I$$

$$A^I = e^{\frac{i}{\hbar}H_0 t} A^S e^{-\frac{i}{\hbar}H_0 t}$$

$$i\hbar \frac{dA^I}{dt} = [A^I, H_0]$$

$$i\hbar \frac{\partial}{\partial t} |\alpha, t\rangle_I = V_I |\alpha, t\rangle_I \qquad V_I = e^{\frac{i}{\hbar}H_0 t} V e^{-\frac{i}{\hbar}H_0 t}$$

$$|\alpha, t\rangle_S = \sum_n C_n(t) e^{-\frac{i}{\hbar}E_n t} |n\rangle \qquad \langle n | V | m \rangle e^{\frac{i}{\hbar}(E_n - E_m)t}$$

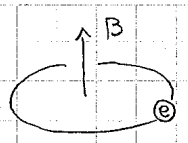
$$|\alpha, t\rangle_I = \sum_n C_n(t) |n\rangle \quad \rightarrow \quad i\hbar \frac{dC_n(t)}{dt} = \sum_m \langle n | V_I | m \rangle C_m(t)$$

2準位問題 $\begin{pmatrix} E_1 & \gamma e^{i\omega t} \\ \gamma e^{-i\omega t} & E_2 \end{pmatrix}$

$$i\hbar \dot{C}_1 = \gamma e^{i(\omega + \omega_{12})t} C_2$$
$$i\hbar \dot{C}_2 = \gamma e^{-i(\omega - \omega_{21})t} C_1$$

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AB effect



$$H = \frac{p^2}{2m} = -\frac{\hbar^2}{2mr^2}\left(\frac{d}{d\theta}\right)^2$$

$$\varphi_n = \frac{1}{\sqrt{2\pi}} e^{in\theta} \qquad E_n = \frac{n^2\hbar^2}{2mr^2}$$

$$\Phi = \int B\,dS = \pi r^2 B = \int A\,dl = 2\pi r A$$

$$A = \frac{\Phi}{2\pi r}$$

$$H = \frac{1}{2m}\left(P - \frac{e}{c}A\right)^2 = -\frac{\hbar^2}{2mr^2}\left(\frac{d}{d\theta} - \frac{ie\Phi}{2\pi\hbar c}\right)^2$$

$$H\left(e^{\frac{ie\Phi}{2\pi\hbar c}\theta}\phi\right) = -\frac{\hbar^2}{2mr^2} e^{\frac{ie\Phi}{2\pi\hbar c}\theta}\left(\frac{d}{d\theta}\right)^2\phi = E\, e^{\frac{ie\Phi}{2\pi\hbar c}\theta}\phi$$

$$\varphi_n^B = \frac{1}{\sqrt{2\pi}} e^{i\left(\frac{\sqrt{2mr^2E}}{\hbar} + \frac{e\Phi}{2\pi\hbar c}\right)\theta}$$

$$\| \quad N$$

$$E_N^B = \frac{\hbar^2}{2mr^2}\left(N - \frac{e\Phi}{2\pi\hbar c}\right)^2$$

$\Phi = \frac{2\pi\hbar c}{e} N'$ 磁束の量子化

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摂動論

$H = H_0 + \lambda V \qquad E_n = E_n^{(0)} + \Delta_n$

$(H_0 + \lambda V)|n\rangle = (E_n^{(0)} + \Delta_n)|n\rangle$

$(E_n^{(0)} - H_0)|n\rangle = (\lambda V - \Delta_n)|n\rangle$

$0 = \langle n^{(0)}|(\lambda V - \Delta_n)|n\rangle = \lambda\langle n^{(0)}|V|n\rangle - \Delta_n \langle n^{(0)}|n\rangle$

$|n\rangle = \frac{1}{E_n^{(0)} - H_0}(\lambda V - \Delta_n)|n\rangle + |n^{(0)}\rangle$

$= \sum_{k\neq n} \frac{1}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle\langle k^{(0)}|(\lambda V - \Delta_n)|n\rangle + |n^{(0)}\rangle$

$= |n^{(0)}\rangle + \sum_{k\neq n} \frac{\lambda\langle k^{(0)}|V|n\rangle - \Delta_n\langle k^{(0)}|n\rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle$

$|n\rangle = |n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \cdots \qquad \Delta_n = \lambda\Delta_n^{(1)} + \lambda^2\Delta_n^{(2)} + \cdots$

λ の1次 $\quad |n^{(1)}\rangle = \sum_{k\neq n} \frac{\langle k^{(0)}|V|n^{(0)}\rangle - \Delta_n^{(1)}\langle k^{(0)}|n^{(0)}\rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle = \sum_{k\neq n} \frac{V_{kn}}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle$

$$\boxed{|n\rangle = |n^{(0)}\rangle + \sum_{k\neq n} \frac{\lambda V_{kn}}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle}$$

$$\boxed{\Delta_n = \lambda\langle n^{(0)}|V|n\rangle = \lambda\langle n^{(0)}|V|n^{(0)}\rangle + \lambda^2 \sum_{k\neq n} \frac{V_{nk}V_{kn}}{E_n^{(0)} - E_k^{(0)}}}$$

check $\quad H|n\rangle = H_0|n^{(0)}\rangle + \lambda V|n^{(0)}\rangle + \sum_{k\neq n} \frac{\lambda V_{kn}}{E_n^{(0)} - E_k^{(0)}} H_0|k^{(0)}\rangle + \sum_{k\neq n} \frac{\lambda V_{kn}}{E_n^{(0)} - E_k^{(0)}} \lambda V|k^{(0)}\rangle$

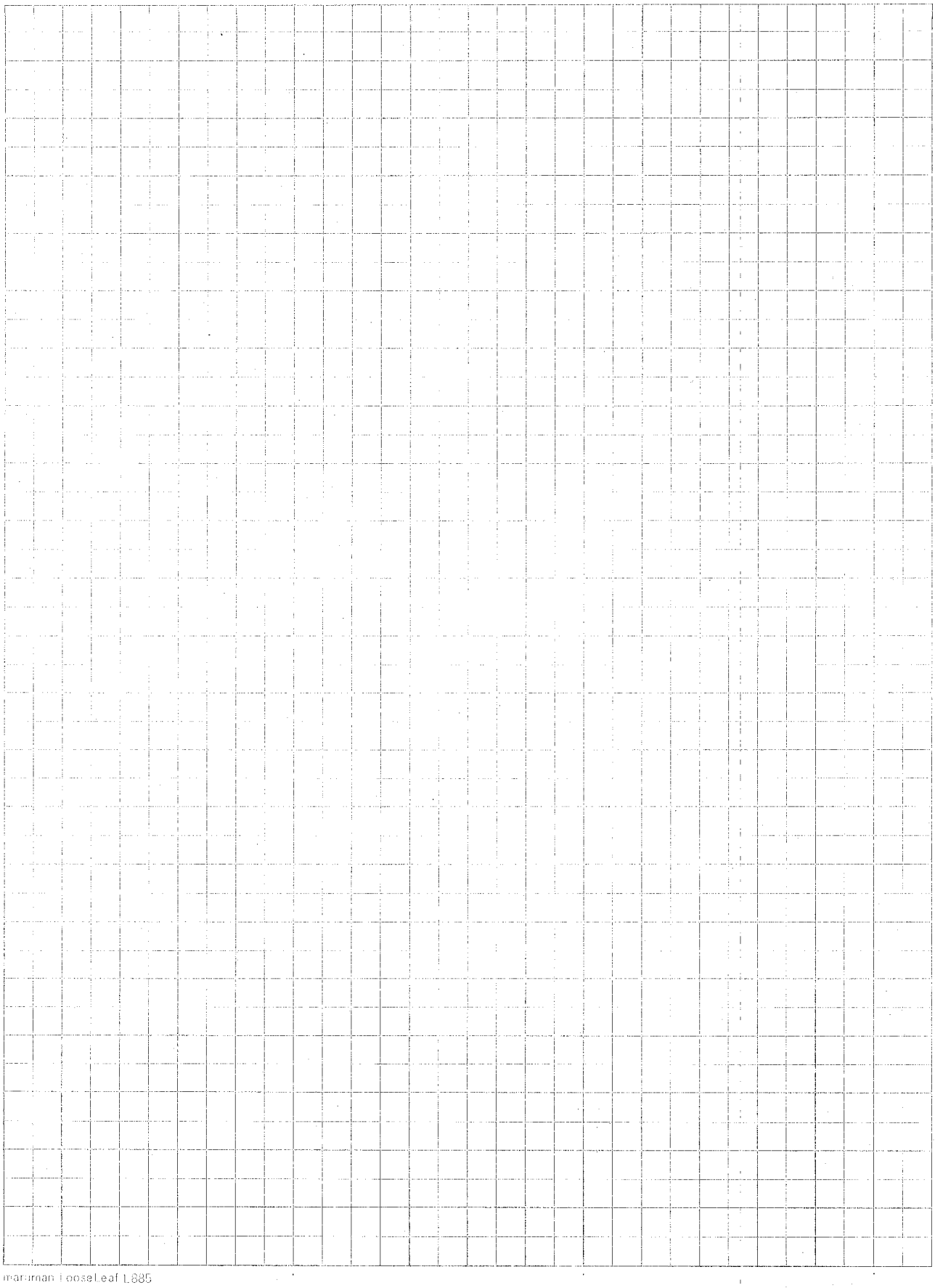
$= E_n^{(0)}|n^{(0)}\rangle + \sum_k \lambda V_{kn}|k^{(0)}\rangle + \sum_{k\neq n} \frac{\lambda V_{kn}E_k^{(0)}}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle + \sum_l \sum_{k\neq n} \frac{\lambda V_{kn}}{E_n^{(0)} - E_k^{(0)}} \lambda V_{lk}|l^{(0)}\rangle$

$= E_n^{(0)}|n^{(0)}\rangle + \lambda V_{nn}|n^{(0)}\rangle$

$+ E_n^{(0)} \sum_{k\neq n} \frac{\lambda V_{kn}}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle + \lambda^2 \sum_{k\neq n} \frac{|V_{nk}|^2}{E_n^{(0)} - E_k^{(0)}} |n^{(0)}\rangle + \sum_{l\neq n} \sum_{k\neq n} \frac{\lambda^2 V_{kn}V_{lk}}{E_n^{(0)} - E_k^{(0)}} |l^{(0)}\rangle$

$= E_n^{(0)}|n\rangle + \lambda\Delta_n^{(1)}|n^{(0)}\rangle + \lambda^2\Delta_n^{(2)}|n^{(0)}\rangle +$

NO.



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多重極展開

$$\frac{1}{|r-r'|} = \frac{1}{r} + \frac{\vec{r}\cdot\vec{r}'}{r^3} + \cdots$$

$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(r')}{|r-r'|}\,d\tau' = \frac{1}{4\pi\epsilon_0 r}\underbrace{\int \rho(r')\,d\tau'}_{Q} + \frac{1}{4\pi\epsilon_0 r^3}\int (\vec{r}\cdot\vec{r}')\,\rho(r')\,d\tau'$$

$$= \frac{Q}{4\pi\epsilon_0 r} + \frac{\vec{r}\cdot\vec{p}}{4\pi\epsilon_0 r^3} \qquad \vec{p} = \int \vec{r}'\rho(r')\,d\tau'$$

$$A(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{J}(r')}{|r-r'|}\,d\tau' = \frac{\mu_0}{4\pi r}\int \vec{J}(r')\,d\tau' + \frac{\mu_0}{4\pi r^3}\int (\vec{r}\cdot\vec{r}')\,\vec{J}(r')\,d\tau'$$

$$= \frac{\mu_0 \vec{m}\times\vec{r}}{4\pi r^3}$$

$$\nabla'\cdot(x'\vec{J}') = J_{x'}' + x'(\nabla'\cdot\vec{J}')$$

$$\int x'\vec{J}'\cdot d\vec{S} = \int J_{x'}'\,d\tau' - \frac{\partial}{\partial t}\int x'\rho'\,d\tau'$$

$\downarrow$ 0

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Maxwell's eq

NO.

$$\nabla \cdot D = \rho \qquad \int D\,dS = \int \rho\,dV$$

$$\nabla \times E + \frac{\partial B}{\partial t} = 0 \qquad \int E\,dl = -\frac{\partial}{\partial t}\int B\,dS$$

$$\nabla \cdot B = 0 \qquad \int B \cdot dS = 0$$

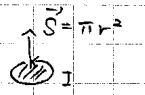
$$\nabla \times H = j + \frac{\partial D}{\partial t} \qquad \int H\,dl = \int j\,dS + \frac{\partial}{\partial t}\int D\,dS$$

$$\nabla \cdot D \times H = j + \frac{\partial \rho}{\partial t} = 0$$

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磁気モーメント

$$\vec{\mu} = \frac{1}{2}\int_V \vec{r}\times\vec{j}\,dV = \frac{I}{2}\int_S \vec{r}\times d\vec{r} = I\vec{S}$$

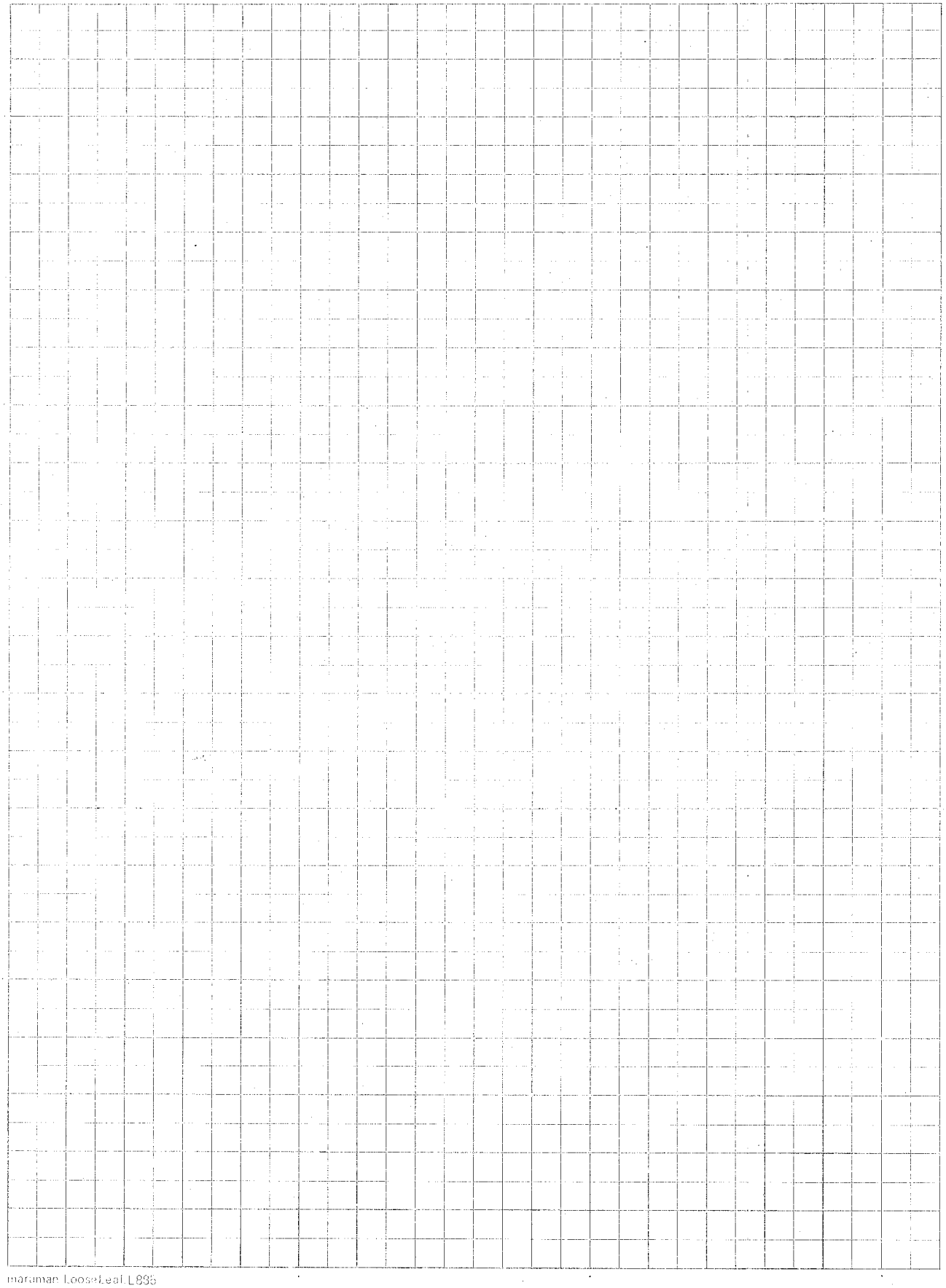


$$\vec{j} = e\vec{v}$$

$$\vec{\mu} = \frac{e}{2m}\,\vec{r}\times\vec{p} = \frac{e}{2m}\,\vec{L} \quad \rightarrow \quad \mu = \frac{e}{2m}(L+2S)$$

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